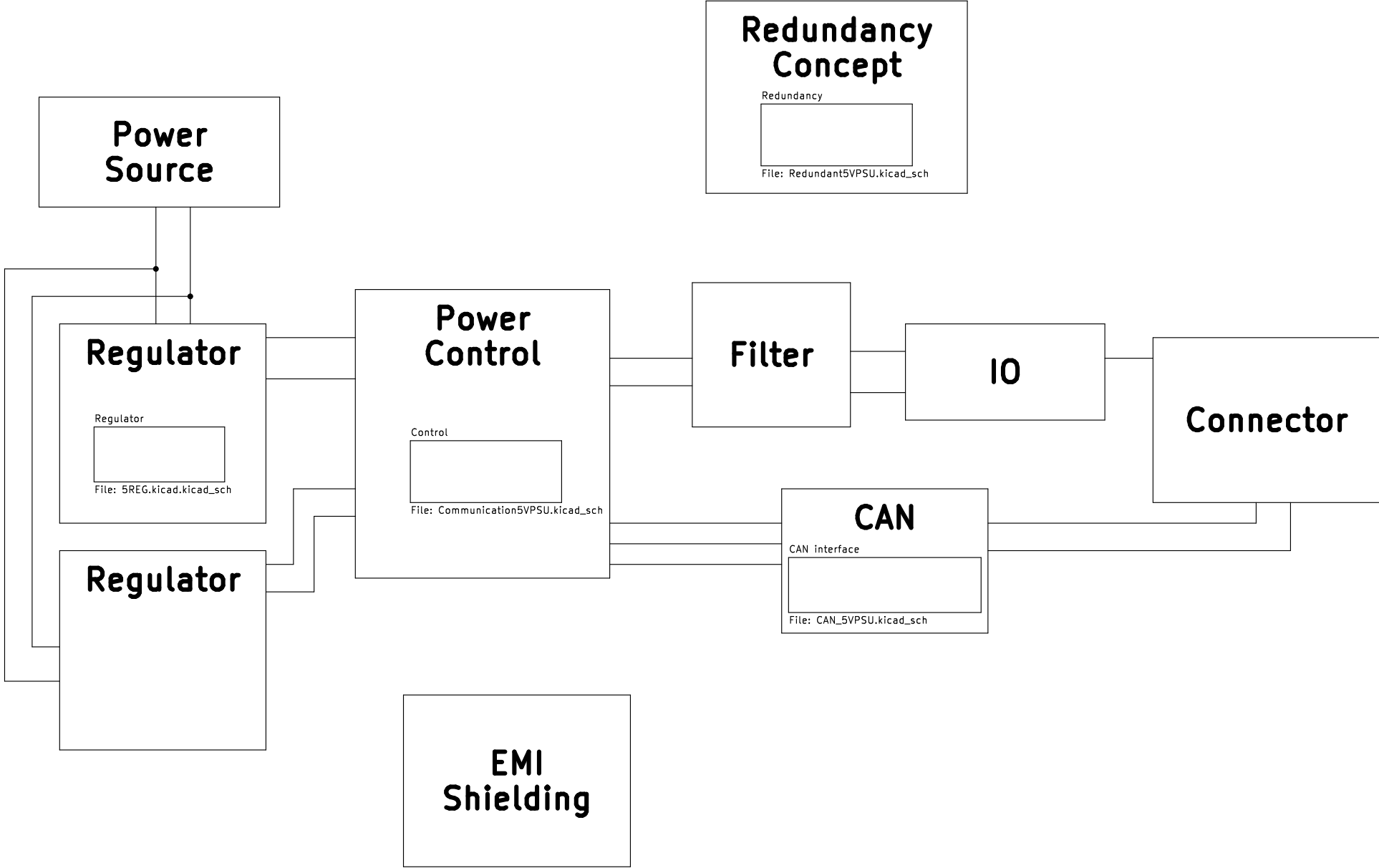
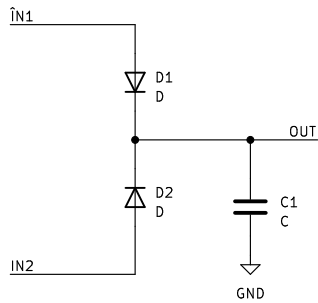


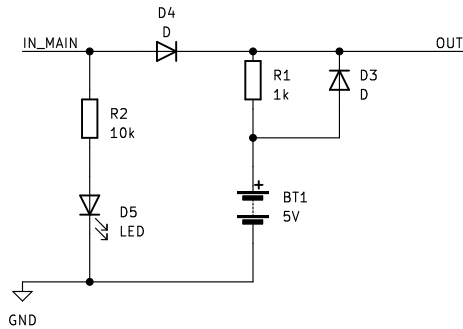
5V Power Module Block Diagram





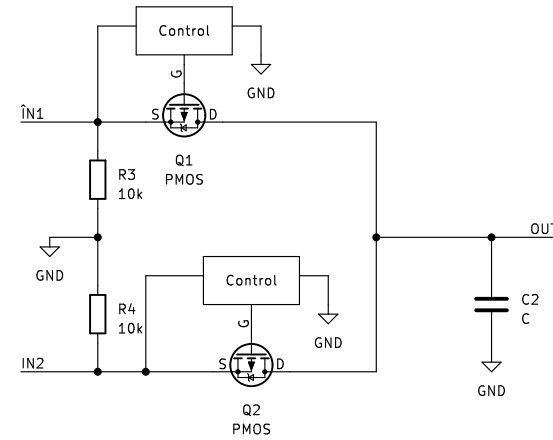
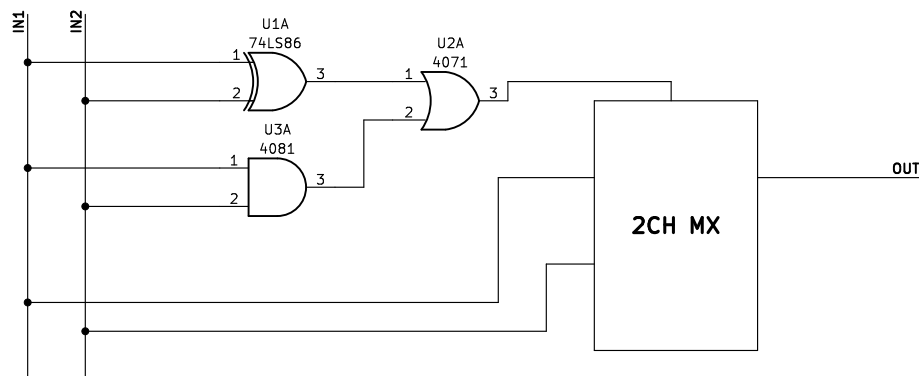
ORing Diodes

Simple, Cheap and Compact.
Protects from Batteryside Faults and Reverse Polarity.
However, high potential for waste heat accumulation
during nominal, stress situations and failures on load side.



Backup Battery

Could alleviate burden on main batteries.
Hardcode solution in case components within PSU
create too much heat or are deemed unreliable.
Can be combined with other redundancy solutions.



ORing MosFET

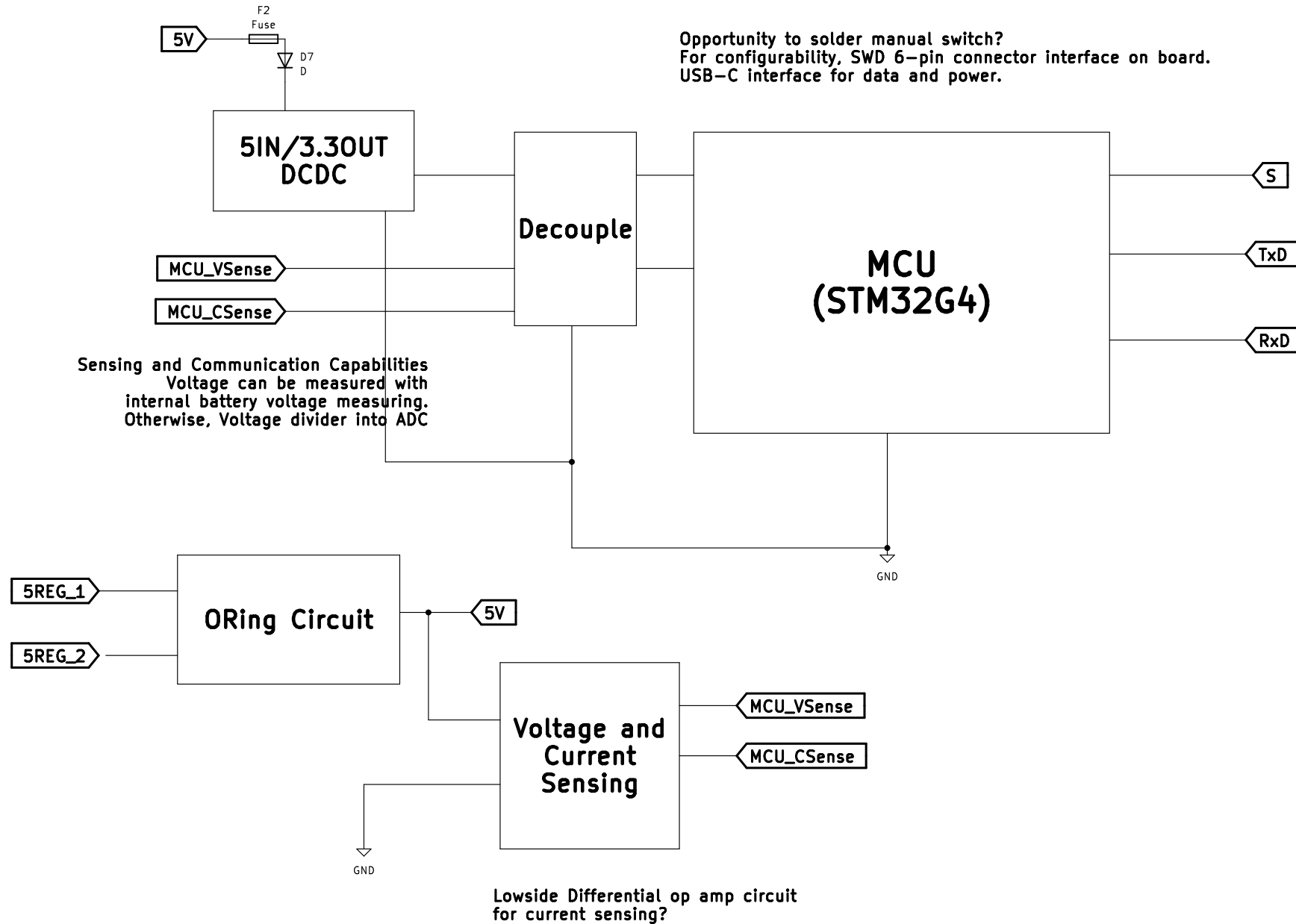
Using MOSFET's instead of diodes reduces waste heat.
However, in need of control signal and no protection from load side.
Additional control module takes up space.

Possible controllers like LT4351 also provide
UV and OV shutoff protection based on hysteresis
with low pass filters to neglect pulses

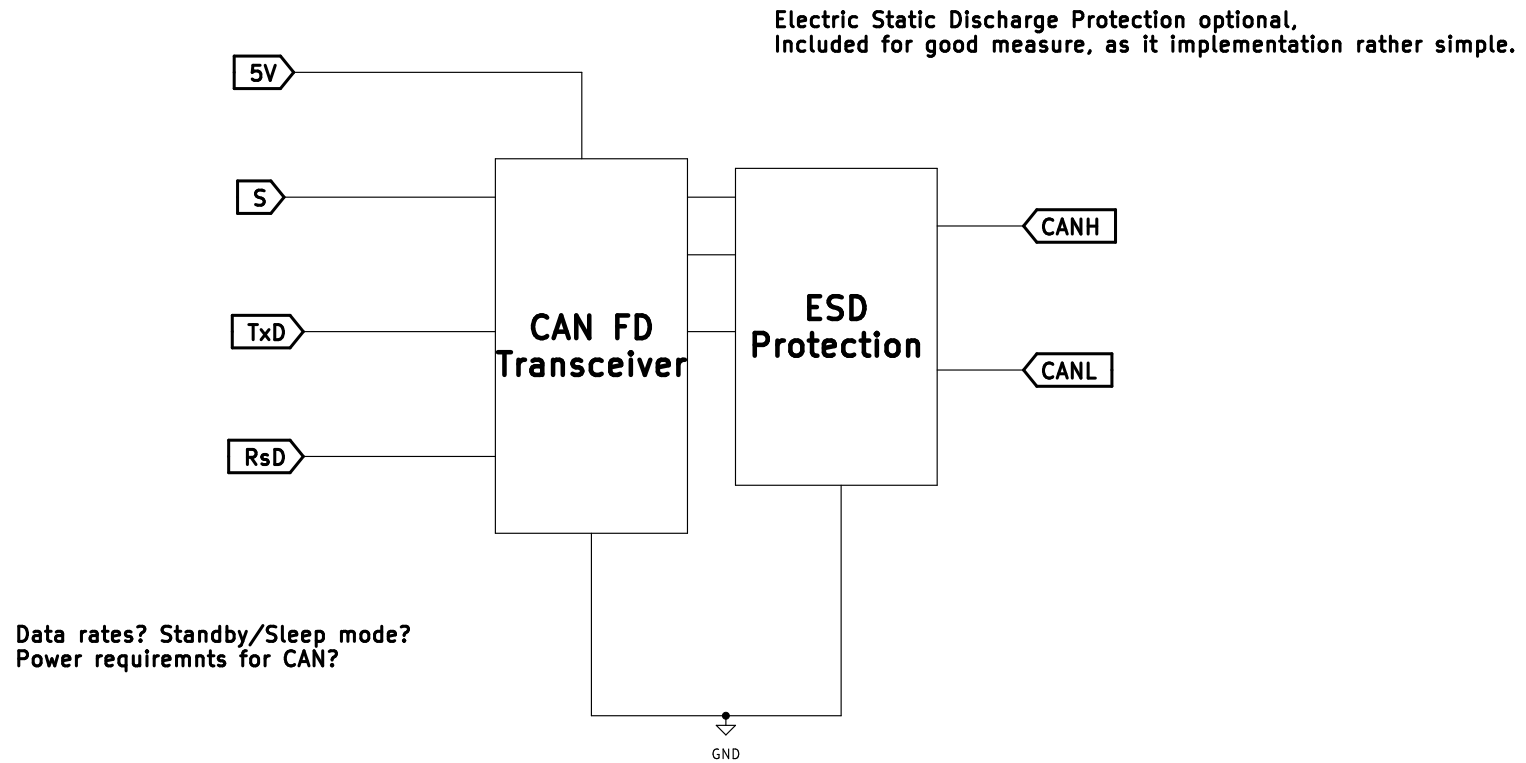


SELECTED

Control Circuit

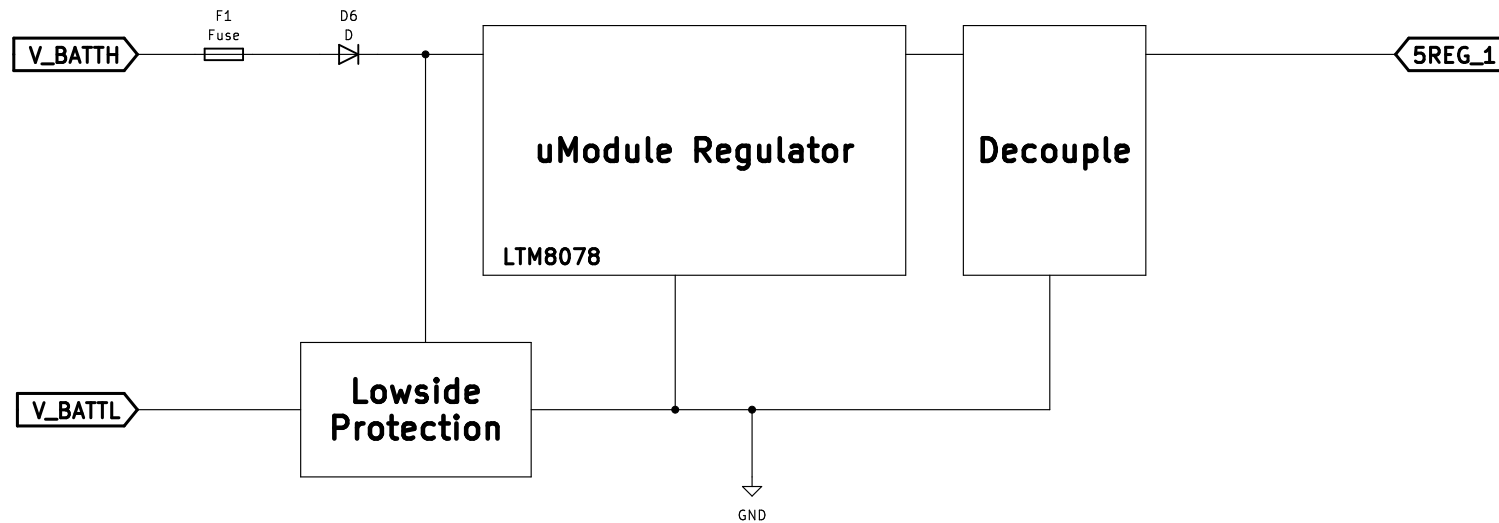


CAN Interface



Regulator Circuit

Max 40V/2.8A support.
Max 2A required, thus fits.
Use dual channel as single bounded output
12CHF per unit. Could find cheaper regs,
but no silent switching.



No built-in EMI shielding, regulators with shielding do exist.
In case, we decide to not implement any shielding,
we could consider buying one with